INDOMETHACIN

Neonatal use: PDA closure

Take with food

CHEMICAL NAME 2-{1-[(4-chlorophenyl)carbonyl]-5-methoxy-2-methyl-1H-indol-3-yl}acetic acid

CLASS Non-selective NSAID (indole acetic acid derivative)

TRADE NAMES Indocin · Indocid · Indomet

THERAPEUTIC USES / INDICATIONS

- Acute gout.
- Rheumatoid arthritis, ankylosing spondylitis, osteoarthritis.
- Closure of patent ductus arteriosus in premature neonates (IV).
- Tocolysis in preterm labour (specialist use, <32 weeks).

ADMINISTRATION

- **PO** capsules with food or milk.
- **PR** suppositories.
- **IV** (neonates) under specialist supervision.

DOSAGES

- **Acute gout:** 50 mg three times daily until pain eases (usually 3–5 days).
- **Arthritis:** 25–50 mg 2–4 times daily; max 200 mg/day.
- **Neonatal PDA:** IV 0.1–0.2 mg/kg per dose for 3 doses, per protocol.

PHARMACOKINETICS

- **Onset:** 30 min. **Peak:** 1–2 h. **Duration:** 4–6 h.
- Metabolised in liver; enterohepatic circulation.
- Excreted in urine and faeces. **t½:** 4.5 h.

PHARMACODYNAMICS

- Potent inhibitor of **COX-1 and COX-2**.
- Strong anti-inflammatory effect; constricts the ductus arteriosus by blocking prostaglandin E_2.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Frontal headache (common), dizziness.
- Nausea, dyspepsia, diarrhoea.
- Drowsiness, confusion in elderly.

ADVERSE EFFECTS (SERIOUS)

- GI ulceration and bleeding.
- Renal impairment; hyperkalaemia; fluid retention.
- Psychiatric effects, depression.
- Neonates: oliguria, NEC, bleeding.

CONTRAINDICATIONS & CAUTIONS

- Active GI ulcer; inflammatory bowel disease.
- Severe renal, hepatic or heart failure.
- NSAID-sensitive asthma.
- Third trimester (except specialist tocolysis); PR use with proctitis.

KEY INTERACTIONS

- **Anticoagulants, steroids:** ↑ bleeding.
- **Lithium, digoxin, aminoglycosides:** ↑ levels.
- **Antihypertensives, diuretics:** ↓ effect.

NURSING MANAGEMENT

- Give with food; watch for headache and dizziness.
- Monitor renal function, K^+ and fluid balance.
- Neonates: monitor urine output, platelets and signs of bleeding.
- Elderly: assess for confusion and falls.
- Assess joint pain and swelling.

PATIENT EDUCATION

- Take with food; report black stools.
- Headache is common; report if severe.
- Avoid driving if dizzy.
- Do not combine with other NSAIDs.

KEY POINTS

- Potent NSAID — used in acute gout and spondylitis.
- IV use to close PDA in premature babies.
- Headache is a typical side effect.
- High GI and CNS side-effect rate.

SOURCES CHECKED **[18]** Indomethacin, oral (US label) **[19]** Indomethacin, IV (US label)

Full references p. 432–457